GETACHEW AMBAYE, Ph.D.

ROBOTICS ENGINEER • INDUSTRIAL AUTOMATION • ROBOT PROGRAMMING & INTEGRATION • EMBEDDED SYSTEMS & AI

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PROFESSIONAL SUMMARY

Robotics Engineer and Ph.D.-trained Industrial Engineer with hands-on experience programming, commissioning, and troubleshooting industrial robotic systems (FANUC, ABB, UR) integrated with PLC/HMI automation and end-of-arm tooling. Combines production-floor deployment experience with applied research in embedded diagnostics, machine learning, and computer vision, giving a full-stack view from mechanical integration through sensing and control to real-time intelligence. Track record spans industry, academia, and peer-reviewed publication, with a portfolio of deployed robotics and automation systems.

TECHNICAL SKILLS

Industrial Robotics: FANUC, Universal Robots (UR), ABB, Robot Programming, Commissioning, Integration & Troubleshooting, ABB RobotStudio, RoboDK, End-of-Arm Tooling (EoAT)

Controls & Automation: Siemens PLC, CLICK PLC, HMI, Industrial Control Systems, Electro-Pneumatics, Hydraulics, Motion Control

Embedded Systems: Raspberry Pi, Microcontrollers, Sensors, IMUs, Data Acquisition Systems (DAQ)

Mechanical & Electrical Drives: AC/DC motor wiring and troubleshooting; low- and high-voltage electrical systems

Programming: Python, MATLAB, C

AI & Computer Vision: Machine Learning, Deep Learning, OpenCV, Image Processing, Computer Vision

CAD, Simulation & Analysis: SolidWorks, CATIA V5, 3D Experience, Autodesk Inventor, AutoCAD, Creo, ANSYS, ABAQUS, MSC ADAMS

Manufacturing: Additive Manufacturing (3D Printing), Rapid Prototyping, Product Development, GD&T

PROFESSIONAL EXPERIENCE

Robotics Engineer II *July 2026 – Present*
Amazon Robotics — Kaleidoscope Innovation

- Commission and troubleshoot industrial robotic systems supporting automation initiatives, reducing downtime and improving cycle efficiency across production lines (Sparrow – fully automated Amazon robotics).

Assistant Professor, Engineering Technology *2025 – 2026*
Austin Peay State University

- Led coursework and applied projects in robotics, machine vision, embedded systems, and automation for students per semester.
- Delivered PLC programming, sensor integration, rapid prototyping, and VR-based instruction, improving lab.

Graduate Research Assistant *2022 – 2025*
Wichita State University

- Conducted research spanning machine learning, robotics, soft robotics, computer vision, and predictive diagnostics, resulting in peer-reviewed publications.
- Performed finite element simulation, experimental validation, and intelligent manufacturing research, improving model prediction accuracy.

Engineering Faculty *2024 – 2025*
Hesston College

- Taught mechatronics, automation, product development, testing, and validation.

SELECTED ROBOTICS & AUTOMATION PROJECTS

Robot Arm Damage Detection

- Developed deep-learning models using vibration signals for robotic structural health monitoring.

Embedded Diagnostic Platform

- Integrated RPM, vibration, displacement, and IMU sensors into a real-time diagnostic platform.

Soft Robot Workspace Prediction Using AI

- Developed machine-learning models to predict soft robotic actuator workspaces using finite element analysis (FEA) data.

Machine Vision Inspection System

- Developed computer vision algorithms for automated drilled-hole misalignment detection.

VR Engineering Training Platform

- Developed immersive virtual reality environments for engineering assembly and visualization.

EDUCATION

Ph.D., Industrial Engineering — Wichita State University, 2025
M.Sc., Mechanical Design Engineering — Jimma University, 2020
B.Sc., Mechanical Engineering — Jimma University, 2016

SELECTED PUBLICATIONS

- Leakage Mechanisms and Airtightness Challenges in FFF-Printed Soft Pneumatic Actuators: A Scoping Review, Electronics, 2026.
- Soft Robot Workspace Estimation via Finite Element Analysis and Machine Learning, Actuators, 2025.
- Robot Arm Damage Detection Using Vibration Data and Deep Learning, Neural Computing and Applications, 2024.
- Soft Robot Design, Manufacturing, and Operation Challenges: A Review, JMMP, 2024.
- Assessing Misalignments in Drilled Holes Using Machine Vision — presented at EAI ICAST 2023, Bahir Dar, Ethiopia (Best Paper Award).
- Detecting Small Defects in Drilled Holes Using Machine Vision — presented at the 5th International Conference on ICT for Development for Africa (ICT4DA), 2023.

CERTIFICATIONS & LICENSES

Engineering, Quality & Manufacturing

- ASME BPV Code, Section V: Nondestructive Examination Overview — ASME, 2025
- ASME GD&T Drawing Interpretation — ASME, 2025
- ASME Boiler & Pressure Vessel Certification Process — ASME, 2026
- Additive Manufacturing Manufacturability: Laser Powder Bed Fusion — ASME, 2026
- IISE Lean Six Sigma Green Belt — Institute of Industrial and Systems Engineers, 2024
- Higher Diploma in Engineering Education

Safety & Compliance

- OSHA 30-Hour Outreach Training — General Industry — 360training, 2026
- OSHA 10-Hour Outreach Training — General Industry — 360training, 2026
- OSHA Focus Four Hazards — 360training, 2026
- NFPA 70E (2024): Standard for Electrical Safety in the Workplace — 360training, 2026 (valid through 2029)
- Lockout/Tagout (LOTO) Certification — 360training, 2026 (valid through 2029)

Software, Data & Automation

- AutoCAD Essentials for Mechanical Engineers — Skill-Lync, 2025
- Microsoft Security Essentials Professional Certificate — Microsoft & LinkedIn Learning, 2025
- Data Science: Working with Data — Alison, 2025
- Computational Fluid Mechanics: Airflow Around a Spoiler — Coursera Project Network, 2025

AWARDS & MEMBERSHIPS

- Outstanding Graduate Student Award (2025)
- Best Paper Award, EAI ICAST (2023)

Professional Memberships: IEEE • ASME • ASEE • ASNT • Alpha Pi Mu